CSI3007	Advanced Python Programming	L	T	P	J	C
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Pre-requisite	CSE1001	**Syllabus version**				
		1.0				

Course Objectives:

1. To be able to apply advanced python programming concepts for industry standard problems.
2. To perform advanced Data Preprocessing tasks like Data Merging and Mugging
3. To be able to develop powerful Web-Apps using Python

Course Outcome:

1. Understand the nuances of Data Structures
2. Derive an understanding of a classes and objects and their potential
3. Gain knowledge of multithreading concepts and implementing the same
4. Appreciate the difference between different data processing techniques
5. Learn to apply Python features for Data Science
6. Get an insight into Metrics Analysis
7. Develop web-apps and build models for IoT

Student Learning Outcomes (SLO): **1, 5, 14**

Module:1	**Data Structures**	**4 Hours**

Problem solving using Python Data Structures : LIST, DICT, TUPLES and SET- Functions and Exceptions – Lamda Functions and Parallel processing – MAPS – Filtering - Itertools – Generators

Module:2	**Classes and Objects**	**4 Hours**

Classes as User Defined Data Type ,Objects as Instances of Classes, Creating Class and Objects, Creating Objects By Passing Values, Variables & Methods in a Class Data Abstraction, Data Hiding, Encapsulation, Modularity, Inheritance, Polymorphism

Module:3	**Multithreading in Python**	**4 Hours**

Python Multithreading and Multiprocessing Multithreading and multiprocessing Basics – Threading module and example – Python multithreading - Multithreaded Priority Queue

Module:4	**Data Processing**	**5 Hours**

Handling CSV, Excel and JSON data - Creating NumPy arrays, Indexing and slicing in NumPy, Downloading and parsing data, Creating multidimensional arrays, NumPy Data types, Array Attribute, Indexing and Slicing, Creating array views copies, Manipulating array shapes I/O – MATPLOT LIB

Module:5	**Data Science Perspectives**	**4 Hours**

Using multilevel series, Series and Data Frames, Grouping, aggregating, Merge DataFrames, Generate summary tables, Group data into logical pieces, Manipulate dates, Creating metrics for analysis

Module:6	**Data Handling Techniques**	**3 Hours**

Data wrangling ,Merging and joining,- Loan Prediction Problem, Data Mugging using Pandas

Module:7	**Web Applications**	**4 Hours**

Web Applications With Python – Django / Flask / Web2Py – Database Programming – NoSQL databases - Embedded Application using IOT Devices - Building a Predictive Model for IOT and Web programming

Module: 8	**Recent Trends**	**2 Hours**
	Total Hours	**30 Hours**

Text Book(s)

1	Doug Farrell, The Well Grounded Python Developer; Manning Publications, 2021
2	Paul Barry, Head-First Python, O-Reilly Media, 2016

Reference Book(s)				
1	Zed A Shaw, Learn Python the Hard Way - A Very Simple Introduction to the Terrifyingly Beautiful World of Computers and Code, Addison Wesley Press, 2013			
2	Eric Mathews, Python Crash Course, Second Edition, No Starch Press, 2019			
3	Michael Kennedy, Talk Python: Building Data-Driven Web Apps with Flask and SQLAlchemy, Manning Publications, 2020			
List of Experiments				**Hours**
1. Working with very large integers/different Data Formats				2 Hour
2. Rewriting an immutable string/String Manipulation				2 Hour
3. Using the Unicode characters that aren't in the keyboard				2 Hour
4. Encoding strings- ASCII and UTF 8				2 Hour
5. Writing list related type hints				4 Hours
6. Building sets with literals, adding, comprehensions and operators				4 Hours
7. Extending a built-in collection – a list that does statistics				4 Hours
8. Using properties for lazy attributes				4 Hours
9. Creating a breadboard prototype Circuit for IoT Program				6 Hours
10. Creating complex structures – maps of lists				6 Hours
11. Using Flask framework for RESTful APIs				6 Hours
12. Implementing authentication for Web Services				6 Hours
13. Application Integration				6 Hours
14. Combining many applications using Command Design Pattern				6 Hours
Total Hours				**60 Hours**
Mode of Evaluation: Project/Activity				
Recommended by Board of Studies		11-02-2021		
Approved by Academic Council	No.61	Date	18-02-2021	